

CHETANKUMAR KHADKE

Staff Software Engineer - LLM & Agentic AI

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PROFESSIONAL SUMMARY

Staff Software Engineer with 13 years in natural language processing and document intelligence, spanning classical ML, transformer-based document models, and production LLM and agentic systems. Began in NLP and ML research at Tata Consultancy Services, mining process models from software test artifacts and building text analytics for the ignio cognitive automation platform, work published as four peer-reviewed IEEE papers. As Engineering Lead at Persistent Systems, built knowledge graphs on Neo4j with custom NER, relation extraction, and entity linking, alongside an intent-classification engine for conversational AI. At Xoriant, designed and trained the custom LayoutLM, Donut, and UDOP models behind an enterprise Document AI platform, improving extraction accuracy by 70%. Now at Xpanse, leading LLM and multi-agent integration (tool calling, Model Context Protocol, and context-aware guardrails) across 500+ document types and millions of documents weekly, for a further ~30% accuracy gain. Certified in Machine Learning (Silver Medal) and Deep Learning by IIT Madras.

TECHNICAL SKILLS

Large Language Models (LLM) & Generative AI: Gemini 3 Pro, Gemini Flash, Claude Sonnet on Amazon Bedrock, Llama 3.2, Mistral AI, OpenAI, Qwen 2.5 VL, LLaVA, Llama-cpp, Hugging Face Transformers, LangChain, LlamaIndex, Fine-Tuning, Prompt Engineering, RAG, Hallucination Mitigation, Guardrails, Model Evaluation

Agentic AI: Multi-Agent Systems, Agent Orchestration, Tool/Function Calling, Model Context Protocol (MCP), Agentic RAG, Human-in-the-Loop, Agent-to-Agent (A2A) Communication, Planning & Reasoning, Memory Management

NLP & Document AI: LayoutLM, Donut, UDOP, BERT, DocVQA, Named Entity Recognition (NER), Named Entity Linking (NEL), Text Classification, Clustering, Topic Modeling, Sentiment Analysis, Embeddings, Vector Search, Semantic Similarity, spaCy, NLTK, Sentence-Transformers, BERTopic, FastText

MLOps & Cloud: AWS (SageMaker, Bedrock, ECR, ECS, S3, CloudWatch, Inference Endpoints), GCP, Docker, MLflow, Git, CI/CD, Model Deployment Pipelines, Containerization, Model Monitoring

Data & Databases: SQL, MySQL, MongoDB, Neo4j, Elasticsearch, Knowledge Graphs

Programming Languages: Python, R, Java

PROFESSIONAL EXPERIENCE

Staff Software Engineer

Feb 2025 - Present

Xpanse

New Jersey, USA

Document AI platform for automated classification and data extraction across 500+ document types and 500+ data fields, processing millions of documents weekly.

- Led LLM integration for the production extraction pipeline, raising end-to-end field extraction accuracy by approximately 30% across 500+ field types.
- Designed and shipped a production multi-agent orchestration layer coordinating specialized classification, extraction, and validation agents, using LLM tool/function calling to invoke internal APIs and the rules engine.
- Built Model Context Protocol (MCP) integrations exposing platform tools and document context to the models through a standard interface, removing bespoke glue code for each new capability.
- Engineered and fine-tuned prompt libraries for Gemini 3 Pro, Gemini Flash, and Claude Sonnet on Amazon Bedrock, delivering precise, context-rich extraction across heterogeneous document layouts.
- Introduced context-aware guardrails and automated validation checks that sharply reduced hallucination rates and produced consistent, auditable outputs on business-critical fields.
- Built a framework that generates field-specific prompts on demand from document-field mappings, significantly cutting onboarding time for new document types.

Xoriant Solutions

May 2021 - Feb 2025

Software Engineer

Aug 2024 - Feb 2025 | New Jersey, USA

Trade document extraction and reconciliation across contracts, invoices, bills of lading, and financial statements, with real-time discrepancy detection.

- Created and tuned context-aware prompt strategies on Gemini 1.5 Pro for high-precision extraction from unstructured trade and financial documents, mitigating hallucination risk on data where errors carry direct monetary cost.

- Designed reconciliation status checks combining LLM reasoning with sentence-embedding similarity matching to detect and report cross-source discrepancies in real time.

Technology Specialist

May 2021 - Jul 2024 | Pune, India

Built the core document-understanding model layer for an enterprise Document AI platform.

- Led the design, training, and evaluation of custom document-understanding models built on LayoutLM, Donut, and UDOP, achieving a 70% improvement in data extraction accuracy over the previous baseline.
- Spearheaded a Named Entity Recognition module, benchmarking multiple NER libraries and custom-trained models, and established the client-specific annotation and labeling workflows behind the training data.
- Implemented Donut-based DocVQA and LLM-driven approaches to extract from unseen and low-template document layouts without per-format rules.
- Owned model deployment pipelines end to end, containerizing models with Docker and shipping to AWS ECR/ECS with MLflow experiment tracking.

Engineering Lead

Aug 2018 - May 2021

Persistent Systems

Pune, India

Knowledge graph construction and conversational AI for enterprise data discovery.

- Led development of a knowledge graph linking organizations, individuals, and content across diverse data sources, modeled and served on Neo4j to enable relationship discovery.
- Built custom NER and relation-extraction classification models, and performed Named Entity Linking against Wikipedia using spaCy to populate and disambiguate graph entities.
- Implemented semantic matching and sentence-similarity algorithms for contextual understanding and entity resolution across sources.
- Developed the intent-classification engine for an automated customer-support chatbot using SVM, Naive Bayes, and RNN models combined with sentence-similarity features.

Systems Engineer to IT Analyst

NOV 2013 - Aug 2018

Tata Consultancy Services

Pune, India

Applied NLP/ML research and delivery for the ignio cognitive automation product and TCS Research.

- Built NLP and time-series pipelines that extracted structured, meaningful information from large-scale IT ticket data, computing key performance metrics and trends for the ignio team.
- Applied clustering, classification, and text-matching algorithms to group similar events and improve similarity-detection accuracy on unstructured text.
- Built a system that mines process and workflow models from system integration test cases, using NLP preprocessing and ML classification to recover software behavior from test artifacts.

EDUCATION

Bachelor of Technology, Computer Science

2009 - 2013

Dharmsinh Desai University, Nadiad, Gujarat, India

CERTIFICATIONS & AWARDS

- Machine Learning (Silver Medal) and Deep Learning - Indian Institute of Technology Madras (NPTEL)
- Bravo Award, received 3 times for outstanding work - Persistent Systems
- XOR-Champ Award for outstanding work - Xoriant Solutions

RESEARCH PUBLICATIONS

1. Khadke C., Rana S., Shah V. "Recovering Workflows from Functional Tests" - IEEE ICPC, Florence, Italy, 2015.
2. Shah V., Khadke C., Rana S. "Mining Process Models and Architectural Components from Test Cases" - IEEE ICSTW, Graz, Austria, 2015.
3. Shah V., Khadke C., Rana S. "Knowledge Transition: Discovering Models from Functional Tests" - IEEE ISSRE, Gaithersburg, MD, USA, 2015.
4. Khadke C. "Knowledge Acquisition (KnowX)" tool demonstration - ENASE, Rome, Italy, 2016.

Full publication list: dblp.uni-trier.de/pers/hd/k/Khadke:Chetan

Technical writing: medium.com/@khadkechetan